

Nuclear Fission and Fusion

Answer key with worked solutions

A. Understanding

1. Definitions.

(a) **Mass defect:** the mass of the separate protons and neutrons minus the mass of the nucleus. **Binding energy:** the energy needed to separate the nucleus into its nucleons, $E = \Delta m c^2$. (b) Binding energy per nucleon is **highest near iron-56**. Fission of a heavy nucleus and fusion of light nuclei both produce nuclei **higher on the curve** (more tightly bound); the increase in total binding energy is released.

B. Nuclear equations

2. Completed equations.

(a) 2_0^1n , **fission** ($236 = 90 + 144 + 2$) (b) $^{141}_{56}\text{Ba}$, **fission** (c) $_0^1\text{n}$, **fusion** (d) $_0^1\text{n}$, **fusion**

C. Mass and energy

3. Helium-4.

$$(a) \Delta m = 2(1.007276 \text{ u}) + 2(1.008665 \text{ u}) - 4.001506 \text{ u} = \mathbf{0.030376 \text{ u}}$$

$$\Delta m = 0.030376 \text{ u} \times 1.66054 \times 10^{-27} \text{ kg/u} = \mathbf{5.044 \times 10^{-29} \text{ kg}}$$

$$(b) E = (5.044 \times 10^{-29} \text{ kg})(2.998 \times 10^8 \text{ m/s})^2 = \mathbf{4.534 \times 10^{-12} \text{ J}}$$

$$(c) \frac{4.534 \times 10^{-12} \text{ J}}{4 \text{ nucleons}} = \mathbf{1.13 \times 10^{-12} \text{ J per nucleon}}$$

4. 1.00 g of matter.

$$E = (1.00 \times 10^{-3} \text{ kg})(2.998 \times 10^8 \text{ m/s})^2 = \mathbf{8.99 \times 10^{13} \text{ J}} \text{ (kg m}^2\text{/s}^2\text{ = J)}$$

D. Fission and reactors

5. Reactor parts.

(a) **Moderator** (water or graphite): slows the neutrons so that uranium-235 captures them more easily. (b) **Control rods** (boron or cadmium): absorb neutrons; raising or lowering them controls the rate of fission. (c) **Coolant**: carries heat away from the core to make steam for the turbines. (d) **Critical mass**: the smallest mass of fissile fuel that can sustain a chain reaction.

6. Fission and fusion.

Fission **splits heavy nuclei**, fusion **joins light** ones; fission produces **long-lived radioactive** waste, fusion mainly helium; fission is used in **power stations** today, fusion is **experimental**. Fusion needs very high temperatures because the nuclei are **positively charged and repel** each other: only very fast collisions bring them close enough to fuse.

7. Energy from uranium-235.

(a) $200 \text{ MeV} \times 1.602 \times 10^{-13} \text{ J/MeV} = 3.204 \times 10^{-11} \text{ J per fission}$

(b) $3.204 \times 10^{-11} \text{ J} \times 6.022 \times 10^{23} \text{ mol}^{-1} = 1.929 \times 10^{13} \text{ J/mol}$

(c) $\frac{1.929 \times 10^{13} \text{ J/mol}}{235.04 \text{ g/mol}} = 8.21 \times 10^{10} \text{ J/g}$

(d) Methane: $\frac{890 \text{ kJ/mol}}{16.04 \text{ g/mol}} = 55.5 \text{ kJ/g} = 5.55 \times 10^4 \text{ J/g}$; ratio = $\frac{8.21 \times 10^{10} \text{ J/g}}{5.55 \times 10^4 \text{ J/g}} \approx 1.48 \times 10^6$ (about one and a half million times)

E. Fusion and the Sun

8. D-D fusion.

$$\Delta m = 2(2.014102 \text{ u}) - (3.016029 \text{ u} + 1.008665 \text{ u}) = 0.003510 \text{ u}$$

$$E = 0.003510 \text{ u} \times 931.5 \text{ MeV/u} = 3.27 \text{ MeV}$$

9. The Sun.

$$(a) \Delta m = \frac{E}{c^2} = \frac{3.83 \times 10^{26} \text{ J}}{(2.998 \times 10^8 \text{ m/s})^2} = \mathbf{4.26 \times 10^9 \text{ kg}} \text{ each second (J = kg m}^2\text{/s}^2\text{, so J } \div \text{ (m}^2\text{/s}^2\text{) = kg)}$$

$$(b) \text{ 1.00 \% of the mass: } 0.0100 \times 1.989 \times 10^{30} \text{ kg} = 1.989 \times 10^{28} \text{ kg}$$

$$t = \frac{1.989 \times 10^{28} \text{ kg}}{4.26 \times 10^9 \text{ kg/s}} = 4.67 \times 10^{18} \text{ s}$$

$$4.67 \times 10^{18} \text{ s} \times \frac{1 \text{ yr}}{3.156 \times 10^7 \text{ s}} = \mathbf{1.48 \times 10^{11} \text{ yr}}$$